

**Decide whether ToT reasoning is appropriate for your agent. (Required)**

- 1. Explain whether ToT reasoning would improve your capstone design or hinder it.**

ToT would improve my agent.

**If ToT is appropriate for your project, complete the following sections:**

- 1. Identify where ToT reasoning will be used within your capstone architecture.**
  - a. Describe the part of your workflow where ToT reasoning will be used.**
  - b. Explain why a linear chain of thought may fail in this area due to premature commitment, high branching, or constraint complexity.**

ToT will be used during the planning phase of my workflow. Traditional CoT would work well if the user already has ideas of what they'd like to plan for. For instance, if the user knows they want a three day trip to WA, the dates planned, the preferred airlines/hotels, and other plans, then this narrow path would be sufficient. However, if the user has a broader scope, such as a \$2000 trip to anywhere in the US, then the ToT would work better. It would create several pathways, compare them to the user's preferences it has learned previously, and prune branches that may fall short.

- 2. Define your ToT structure.**

- a. Specify how reasoning will be represented as a tree.**

- 1. Include:**

- i. What a "thought" represents in your system.**

- i. A thought represents one potential item in the itinerary. This includes flight, dates, hotels, activities, scheduling, etc.

- ii. What constitutes a node, a branch, and depth.**

- i. A branch would be the overall itinerary. Each item in there would be a node. The depth would be the total items in the itinerary.

- iii. What the expected branching factor and depth limit are.**

- i. Given that ToT will be primarily used as an exploratory tool, the branching factors will be around 10. The depth will be around 20. Since

each node is an item, it has to include things such as every meal, activities to do, etc. This has to be for each day. It may change depending on the number of days the trip is.

**iv. When the system terminates and selects a final output.**

- i. The final output is chosen when all nodes fit within the user's constraints and previously learned preferences.

**3. Describe your search and evaluation strategy.**

**a. Choose one primary search strategy and explain why it fits your use case:**

1. DFS will be used because exploring a single path is the least computational and logical when it comes to figuring out an itinerary. Once the agent comes across a bad node, it'll just go back one item and explore a new path.
2. For example, if a user wants to explore the east coast but has a tight budget, it might start with NYC. After exploring, it realizes that NYC is too expensive for the budget, it will prune and explore a different city.

**b. Explain how candidate branches will be assessed. Include:**

**1. The evaluation criteria or scoring rubric. Who performs the evaluation (the critic agent, heuristic checks, tool calls, or a combination).**

- i. There will be a combination. The critic agent and heuristic will be used for more subjective statements such as the requirement of "the place must be less touristy and secluded". There's no quantitative measure for it. For other constraints such as budget, a simple tool call to compare the prices can be used.
- ii. These requirements are defined in real time by the user or past preferences learned.

**2. Pruning thresholds and failure conditions.**

- i. The user will be providing requirements for the trip. If at any point the item contradicts any of these, that pathway will be pruned. For instance, if the total budget is \$500, if a path reaches over that threshold, it will be pruned.

**3. How ties are resolved (voting, reranking, or selection policy).**

- i. When there's a tie after reviewing all the requirements and user preferences, a simple RNG will be used to determine the outcome. If the situation occurs at the end of a branch depth, the options will be presented to the user. Based on the feedback, their preferences will be updated accordingly.

**4. How the strategy will be constrained by compute, latency, or cost.**

- i. This task is a relatively low compute compared to other agents. Based on the user's preferences/settings, the agent will explore different level depths and branching factors to fit the user's budget/time to use the tool.

**4. Map ToT roles to implementation tools.**

**a. At a minimum, include:**

**1. Thought generator.**

- i. This is the core of the agent. It will use the thought generator to find and search for applicable itinerary items.

**2. Critic or evaluator.**

- i. This will evaluate each item in accordance to the user's requirements and preferences.

**3. Decision maker or controller.**

- i. This will decide which path to explore and which to prune.

**4. Memory or state manager.**

- i. This won't have a large factor given that DFS will be used.

**b. For each role, identify the implementation tool you plan to use and explain why:**

**1. CrewAI (agent definition, task routing, role separation).**

- i. CrewAI will be used to create a subagent for each of the ToT roles. Additionally, a subagent for each core component of the itinerary planning will be created as well such as flight subagent, hotel subagent, rental car, etc.

**2. LangChain (control flow, LCEL, tool integration, search orchestration).**

- i. This will be used as the main orchestrator for creating the structure. It will define what constitutes good, what gets pruned, what moves on, etc.

**3. MCP (shared state, context passing, branch tracking).**

- i. MCP servers will be used to access live data, such as flights and hotels, and be called via the agents defined in CrewAI.

**5. Identify one risk and one mitigation.**

**a. Choose one risk relevant to your agent, such as:**

- 1. Branch explosion and high computational cost.**
- 2. Weak evaluation signals leading to pruning the best branch.**
- 3. Overexploitation that increases latency without accuracy gains.**
- 4. Inconsistent state management across branches.**

**b. Describe one mitigation strategy you will apply.**

One risk is that given a new user, it may have really poor evaluation signals. Without any guidance, it will believe every path is good. To combat this, a general survey of each new member will be conducted. It will include things such as preferred airlines, hotels, hobbies, window vs aisle, cuisine, etc. These preferences will then be stored in the user's settings and be utilized to create the evaluation.